Constructing the rank interval with sign changes

Human Compression project: F006 recovery

Author working proof; distinct review pending

Use E_n and $W_n = E_n \rtimes S_n$ from the companion note *Even sign changes and the rank of a string*. The permutation subgroup S_n acts without sign changes. The vector $e_i + e_j$ flips the two signed coordinates i, j. We use the standard end criterion: a string of involutions is a C-marking if its two end deletions are C-markings and their groups intersect in the common middle group. This is the retained McMullen–Schulte criterion, Proposition 2E16.

Lemma 1 (Adjoining a kernel involution)**.** *Let $(s_1, \dots, s_d)$ be a string C-marking of a complement S in a split extension $N \rtimes S$. Let $x \in N$ be a nonidentity involution commuting with $s_1, \dots, s_{d-1}$. Then $(s_1, \dots, s_d, x)$ is a string C-marking of its generated subgroup.*

Proof. Projection to S shows that, for every k,

$$S \cap \langle s_k, \dots, s_d, x \rangle = \langle s_k, \dots, s_d \rangle.$$

Indeed the projected image is the group on the right, and projection is the identity on S; the reverse containment is explicit. The pair (s_d, x) consists of distinct nonidentity involutions and has the intersection property. Induct backwards on k. One end of $(s_k, \dots, s_d, x)$ is an old C-submarking in S; the other is the new suffix already proved C. The displayed identity makes their intersection precisely $\langle s_{k+1}, \dots, s_d \rangle$. The end criterion proves the induction. The case $d = 1$ is the initial pair. Generation of the whole extension is a separate question. □

This proves the intersection step of Chen–Rowley, Lemma 18, for an arbitrary split kernel. No assertion about intersections of projected subgroups is being substituted for the actual intersection property.

1 A symmetric marking with an alternating path at its end

For $n \geq 5$ and $3 \leq d \leq n-2$, put

$$a_i = (i\ i+1)\ (1 \leq i \leq d-2), \qquad b = \prod_{j=d-1,d+1,\dots<n} (j\ j+1), \qquad c = \prod_{j=d,d+2,\dots<n} (j\ j+1).$$

Lemma 2. *The tuple $(a_1, \dots, a_{d-2}, b, c)$ is a string C-marking of S_n with adjacent orders $[3^{d-3}, 6, n-d+2]$.*

Proof. The union of the two matchings b, c is the path on $\{d-1, \dots, n\}$. Its product is a cycle of length $n-d+2$, as is seen by following alternate edges from an endpoint. Thus $\langle b, c \rangle$ is transitive on that set. Conjugating the bridge $(d-2\ d-1)$ by this group gives the transpositions joining $d-2$ to every point of the set, and hence its full symmetric group together with $d-2$. The preceding bridges then give S_n.

For the intersection property first take $d = 3$. Write $a = (1\ 2)$, $b = (2\ 3)t$, where $t = (4\ 5)(6\ 7)\cdots$ is nontrivial and disjoint from $\{1, 2, 3\}$. Since $(ab)^3 = t$,

$$H = \langle a, b \rangle = S_{\{1,2,3\}} \times \langle t \rangle.$$

The subgroup $K = \langle b, c\rangle$ is the dihedral action on the $(n-1)$-cycle given by bc, and fixes 1. Therefore $H \cap K \le H_1 = \langle b, t\rangle$. The points $2, 3$ are adjacent in that cycle and are both fixed by t. A nonidentity dihedral element cannot fix two adjacent cycle points when the length is at least four: a rotation fixes none, and the equations for a reflection fixing successive residues would give $2 = 0$ modulo the cycle length. Thus $t \notin K$ and $H \cap K = \langle b\rangle$. The end criterion gives the rank-three intersection property.

For $d > 3$, deleting a_1 gives the same construction on $\{2, \dots, n\}$ with rank $d-1$. Its first mark is the pure transposition (2 3), and every other mark fixes 2. Apply the acquired pure-leaf corollary of *Restriction kernels in extensions of string markings*, adjoining (1 2) at a new point. That corollary includes the complete intersection proof, with no transitivity assumption on a projected action. Induction proves C. Finally the products of consecutive pure transpositions have order three; $a_{d-2}b$ is a three-cycle times a nonempty disjoint matching, of order six; and the order of bc was computed above. All other required pairs commute directly from their supports. □

This is the explicit family attributed by Chen–Rowley, Theorem 17, to Fernandes–Leemans (2011), Lemma 21. Its proof here uses the already acquired restriction-kernel theorem, with the base intersection checked directly.

2 Odd degree

Theorem 3. *If $n \ge 5$ is odd, then W_n has a rank-n C-marking of type $[4, 3^{n-2}]$ and C-markings in every rank from three through n.*

Proof. Take $v = e_2 + \cdots + e_n \in E_n$ and $s_i = (i\ i+1)$. The adjacent transpositions form the usual C-marking of S_n: every subset generates the independent symmetric groups on its path components, whose intersections are the components of the intersection of the two edge sets. The vector v commutes with s_i for $i \ge 2$. Lemma 1, with the ordering reversed, proves C for $(v, s_1, \dots, s_{n-1})$. Its group contains S_n and the nonconstant vector v, so the module lemma gives all of E_n. It is W_n. The square of vs_1 is $v + s_1 v = e_1 + e_2 \ne 0$, so its order is four; the other adjacent orders are three.

Apply the acquired Brooksbank–Leemans contraction theorem, in *Contraction of string markings through a central factor*, Theorem 3 and Corollary 5. Every bond tested in successive contractions, starting at the third bond, has order three. Each contraction preserves the group and the intersection property, and iteration ends at rank three. This gives precisely the stated interval. □

3 Even degree

Theorem 4. *If $n \ge 6$ is even, then W_n has C-markings in every rank from three through $n-1$. In rank three the type is $[12, n-1]$; in rank $4 \le r \le n-1$ it is $[3^{r-4}, 6, n-r+3, 4]$.*

Proof. For $r \ge 4$ use Lemma 2 with $d = r-1$. Let $(u\ u+1)$ be the last transposition in b and put $v = e_u + e_{u+1}$. This is $(n-1\ n)$ when r is odd and $(n-2\ n-1)$ when r is even. In either case $u > r-2$. Thus v commutes with all pure marks and with b. Append v after c and apply Lemma 1. The group contains S_n and a nonconstant vector, hence is W_n. The matching c does not preserve $\{u, u+1\}$, so $(cv)^2 = v + cv$ is a nonzero vector of order two. This gives the last bond order four and proves the asserted type.

For rank three put

$$a = (e_{n-1} + e_n)(1\ 2), \quad b = (2\ 3)(4\ 5)\cdots(n-2\ n-1), \quad c = (3\ 4)(5\ 6)\cdots(n-1\ n).$$

These are involutions. The supports of (1 2) and c are disjoint, and c preserves the sign-change pair $\{n-1, n\}$, so a, c commute. On signed letters $q = ab$ has a three-cycle on $\{1, 2, 3\}$ and its negative copy, a four-cycle on $\{\pm(n-2), \pm(n-1)\}$, a sign flip at n, and the remaining disjoint unsigned transpositions. Hence $|q| = 12$ and $q^6 = e_{n-2} + e_{n-1}$. Also bc is a cycle of length $n-1$ on $\{2, \dots, n\}$ and its negative copy.

The projected group contains (1 2) and the transitive group $\langle b, c\rangle$ on the other letters. Conjugates of (1 2) give all the star transpositions, so the image is S_n. The kernel contains the nonconstant q^6. It is invariant under this full image; by the module lemma it is E_n. Generation follows.

For the actual end intersection, every element of $A = \langle a, b\rangle$ has the unique form $b^\epsilon q^k$, $\epsilon \in \{0, 1\}$, $0 \le k < 12$. Since b is unsigned, such an element is unsigned exactly when q^k is unsigned, which the signed four-cycle shows to be equivalent to $4 \mid k$. Since b fixes 1, the same element fixes 1 exactly when $3 \mid k$, by the three-cycle. Therefore

$$A \cap S_n \cap \mathrm{Stab}(1) = \langle b\rangle.$$

As $\langle b, c\rangle$ is contained in the middle group on the left, $\langle a, b\rangle \cap \langle b, c\rangle = \langle b\rangle$. Both end pairs are C, so the end criterion proves the full rank-three assertion. □

The last argument corrects the displayed subgroup calculation in Chen–Rowley, Lemma 19. Both subgroups in its displayed last intersection contain t_1, although t_1 is not unsigned and the claimed intersection is $\langle t_2\rangle$. The actual intersection needed by the lemma is indeed $\langle t_2\rangle$, as the explicit calculation above proves. The rank-three theorem itself is recovered.

Combining Theorems 3 and 4 with the companion upper bound proves the full principal D_n rank conclusions of source Theorems 1 and 2. All constructions, types and the contraction method remain attributed to their original authors. The reorganization uses one kernel-extension lemma and the acquired module, restriction-kernel and contraction arguments. Full dependency proofs and their retained inputs remain required. The exceptional census and distinct review are not completed by this working note; no whole-manuscript or numerical compression claim is attached to it.